

Power BI Assignment 2 - DAX & Data Visualization

E-Commerce Sales Analysis:

In This Assignment, I Analyzed E-commerce sales data using Power BI and DAX. Using Datasets List of orders, Order details, Sales Target. I imported and connected multiple datasets, created calculated columns and measures such as revenue, order count, and YTD sales, and developed visualizations to analyze sales trend, targets, profit, quantity, and regional performance. This project helped me improve my skills in Data modeling, DAX Formulas, KPI, charts and dashboard Visualization.

Queries [3]

Table.SelectRows(#"Filtered Rows", each not List.IsEmpty(List.RemoveMatchingItems(Record.FieldValues

Order ID	Order Date	CustomerName	State	City
B-25601	01-04-2018	Bharat	Gujarat	Ahmedabad
B-25602	01-04-2018	Pearl	Maharashtra	Pune
B-25603	03-04-2018	Jahan	Madhya Pradesh	Bhopal
B-25604	03-04-2018	Divsha	Rajasthan	Jaipur
B-25605	05-04-2018	Kasheen	West Bengal	Kolkata
B-25606	06-04-2018	Hazel	Karnataka	Bangalore
B-25607	06-04-2018	Sonakshi	Jammu and Kashmir	Kashmir
B-25608	08-04-2018	Aarushi	Tamil Nadu	Chennai
B-25609	09-04-2018	Jitesh	Uttar Pradesh	Lucknow
B-25610	09-04-2018	Yogesh	Bihar	Patna
B-25611	11-04-2018	Anita	Kerala	Thiruvananthapuram
B-25612	12-04-2018	Shrichand	Punjab	Chandigarh
B-25613	12-04-2018	Mukesh	Haryana	Chandigarh
B-25614	13-04-2018	Vandana	Himachal Pradesh	Simla
B-25615	15-04-2018	Bhavna	Sikkim	Gangtok
B-25616	15-04-2018	Kanak	Goa	Goa
B-25617	17-04-2018	Sagar	Nagaland	Kohima
B-25618	18-04-2018	Manju	Andhra Pradesh	Hyderabad

5 COLUMNS, 500 ROWS Column profiling based on top 1000 rows

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Queries [3]

Table.RemoveRowsWithErrors(#"Removed Blank Rows", {"Order ID"})

Order ID	Amount	Profit	Quantity	Category
B-25601	1275	-1148	7	Furniture
B-25601	66	-12	5	Clothing
B-25601	8	-2	3	Clothing
B-25601	80	-56	4	Electronics
B-25602	168	-111	2	Electronics
B-25602	424	-272	5	Electronics
B-25602	2617	1151	4	Electronics
B-25602	561	212	3	Clothing
B-25602	119	-5	8	Clothing
B-25603	1355	-60	5	Clothing
B-25603	24	-30	1	Furniture
B-25603	193	-166	3	Clothing
B-25603	180	5	3	Clothing
B-25603	116	16	4	Clothing
B-25603	107	36	6	Clothing
B-25603	12	1	2	Clothing
B-25603	38	18	1	Clothing
B-25604	65	17	2	Clothing

6 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

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Calculated Columns:

Untitled - Power BI Desktop

Search

Sign in

File Home Help Table tools Column tools

Name: Category Type

Format: Text

Summarization: Don't summarize

Data type: Text

Formatting: \$ % -00 Auto

Data category: Uncategorized

Sort by column

Data groups

Manage relationships

New column

Structure

Formatting

Properties

Sort

Groups

Relationships

Calculations

1 Category Type = 'Order Details (1)'[Category] & " - " & 'Order Details (1)'[Sub-Category]

Data

Search

List of Orders (1)

Order Details (1)

Amount

Category

Category Type

Order ID

Profit

Quantity

Sub-Category

Sales target (1)

Order ID	Amount	Profit	Quantity	Category	Sub-Category	Category Type
B-25623	53	1	4	Clothing	Stole	Clothing - Stole
B-25627	55	-39	4	Clothing	Stole	Clothing - Stole
B-25630	133	12	5	Clothing	Stole	Clothing - Stole
B-25649	27	-25	2	Clothing	Stole	Clothing - Stole
B-25670	74	29	3	Clothing	Stole	Clothing - Stole
B-25672	27	-15	1	Clothing	Stole	Clothing - Stole
B-25687	17	6	1	Clothing	Stole	Clothing - Stole
B-25706	31	-11	4	Clothing	Stole	Clothing - Stole
B-25707	8	-6	1	Clothing	Stole	Clothing - Stole
B-25719	29	10	2	Clothing	Stole	Clothing - Stole
B-25722	48	-8	8	Clothing	Stole	Clothing - Stole
B-25723	26	-24	1	Clothing	Stole	Clothing - Stole
B-25732	16	-5	2	Clothing	Stole	Clothing - Stole
B-25733	43	-43	7	Clothing	Stole	Clothing - Stole
B-25740	40	-37	3	Clothing	Stole	Clothing - Stole
B-25745	44	-8	3	Clothing	Stole	Clothing - Stole
B-25758	8	-2	1	Clothing	Stole	Clothing - Stole
B-25765	139	14	3	Clothing	Stole	Clothing - Stole
B-25775	50	-17	2	Clothing	Stole	Clothing - Stole
B-25783	25	-11	1	Clothing	Stole	Clothing - Stole
B-25788	12	3	1	Clothing	Stole	Clothing - Stole

Table: Order Details (1) (500 rows) Column: Category Type (17 distinct values)

Dax & Visualization • Last saved: Today at 8:33 PM

Search

Sign in

File Home Help Table tools Column tools

Name: Revenue Per Order

Format: Whole number

Summarization: Sum

Data type: Whole number

Formatting: \$ % -00 0

Data category: Uncategorized

Sort by column

Data groups

Manage relationships

New column

Structure

Formatting

Properties

Sort

Groups

Relationships

Calculations

1 Revenue Per Order = 'Order Details (1)'[Amount]*'Order Details (1)'[Quantity]

Data

Search

List of Orders (1)

Order Details (1)

Amount

Category

Category Type

Order ID

Profit

Quantity

Revenue Per Order

Sub-Category

Sales target (1)

Order ID	Amount	Profit	Quantity	Category	Sub-Category	Category Type	Revenue Per Order
B-25623	53	1	4	Clothing	Stole	Clothing - Stole	212
B-25627	55	-39	4	Clothing	Stole	Clothing - Stole	220
B-25630	133	12	5	Clothing	Stole	Clothing - Stole	665
B-25649	27	-25	2	Clothing	Stole	Clothing - Stole	54
B-25670	74	29	3	Clothing	Stole	Clothing - Stole	222
B-25672	27	-15	1	Clothing	Stole	Clothing - Stole	27
B-25687	17	6	1	Clothing	Stole	Clothing - Stole	17
B-25706	31	-11	4	Clothing	Stole	Clothing - Stole	124
B-25707	8	-6	1	Clothing	Stole	Clothing - Stole	8
B-25719	29	10	2	Clothing	Stole	Clothing - Stole	58
B-25722	48	-8	8	Clothing	Stole	Clothing - Stole	384
B-25723	26	-24	1	Clothing	Stole	Clothing - Stole	26
B-25732	16	-5	2	Clothing	Stole	Clothing - Stole	32
B-25733	43	-43	7	Clothing	Stole	Clothing - Stole	301
B-25740	40	-37	3	Clothing	Stole	Clothing - Stole	120
B-25745	44	-8	3	Clothing	Stole	Clothing - Stole	132
B-25758	8	-2	1	Clothing	Stole	Clothing - Stole	8
B-25765	139	14	3	Clothing	Stole	Clothing - Stole	417
B-25775	50	-17	2	Clothing	Stole	Clothing - Stole	100
B-25783	25	-11	1	Clothing	Stole	Clothing - Stole	25
B-25788	12	3	1	Clothing	Stole	Clothing - Stole	12

Table: Order Details (1) (500 rows) Column: Revenue Per Order (384 distinct values)

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Name Sales Category Format Text Summarization Don't summarize Data category Uncategorized

Structure Formatting Properties

1 Sales Category = IF('Order Details (1)'[Amount]>AVERAGE('Order Details (1)'[Amount]),"Above Average","Below Average")

Order ID	Amount	Profit	Quantity	Category	Sub-Category	Category Type	Revenue Per Order	Sales Category
B-25623	53	1	4	Clothing	Stole	Clothing - Stole	212	Below Average
B-25627	55	-39	4	Clothing	Stole	Clothing - Stole	220	Below Average
B-25630	133	12	5	Clothing	Stole	Clothing - Stole	665	Below Average
B-25649	27	-25	2	Clothing	Stole	Clothing - Stole	54	Below Average
B-25670	74	29	3	Clothing	Stole	Clothing - Stole	222	Below Average
B-25672	27	-15	1	Clothing	Stole	Clothing - Stole	27	Below Average
B-25687	17	6	1	Clothing	Stole	Clothing - Stole	17	Below Average
B-25706	31	-11	4	Clothing	Stole	Clothing - Stole	124	Below Average
B-25707	8	-6	1	Clothing	Stole	Clothing - Stole	8	Below Average
B-25719	29	10	2	Clothing	Stole	Clothing - Stole	58	Below Average
B-25722	48	-8	8	Clothing	Stole	Clothing - Stole	384	Below Average
B-25723	26	-24	1	Clothing	Stole	Clothing - Stole	26	Below Average
B-25732	16	-5	2	Clothing	Stole	Clothing - Stole	32	Below Average
B-25733	43	-43	7	Clothing	Stole	Clothing - Stole	301	Below Average
B-25740	40	-37	3	Clothing	Stole	Clothing - Stole	120	Below Average
B-25745	44	-8	3	Clothing	Stole	Clothing - Stole	132	Below Average
B-25758	8	-2	1	Clothing	Stole	Clothing - Stole	8	Below Average
B-25765	139	14	3	Clothing	Stole	Clothing - Stole	417	Below Average
B-25775	50	-17	2	Clothing	Stole	Clothing - Stole	100	Below Average
B-25783	25	-11	1	Clothing	Stole	Clothing - Stole	25	Below Average
B-25788	12	3	1	Clothing	Stole	Clothing - Stole	12	Below Average

Table: Order Details (1) (500 rows) Column: Sales Category (2 distinct values)

Dax & Visualization • Last saved: Today at 11:14 AM

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Clipboard

1 Max profit

Order ID Amount Profit

B-25602 561

B-25602 119

B-25603 193

B-25604 157

B-25605 75

B-25609 25

B-25610 43

B-25611 160

B-25613 1603

B-25619 353

B-25622 534

B-25623 149

B-25625 635

B-25628 24

B-25633 711

B-25635 382

B-25636 637

B-25640 122

B-25646 20

B-25647 42

B-25648 55

Table: Order Details (1,500 rows) Column: Max profit

Manage relationships

+ New relationship Autodetect

From: Order Details (Order ID) Relationship To: table (column) Status

Order Details (Order ID) * 1 List of Orders (Order ID) Active

Close

Dax & Visualization • Last saved: Today at 11:14 AM

File Home Help

Clipboard

Excel workbook Enter data OneLake catalog Datasource

Get data SQL Server Recent sources

Transform Refresh data

Manage relationships

New measure column

New table

Calculation group

New parameter

Manage roles View as

Q&A setup

Language Linguistic schema

Q&A

Sensitivity

Public

Share

Data

Tables Model

Search

List of Orders

Order Details

Sales target

City CustomerName Order Date Order ID State

Amount Category Category Type Order ID Profit Quantity

Category Month of Order Date Target

List of Orders

Order Details

Amount

Average Profit Delhi

Category

Category Type

Max profit

Minimum Target

Order Count

Order ID

Profit

Quantity

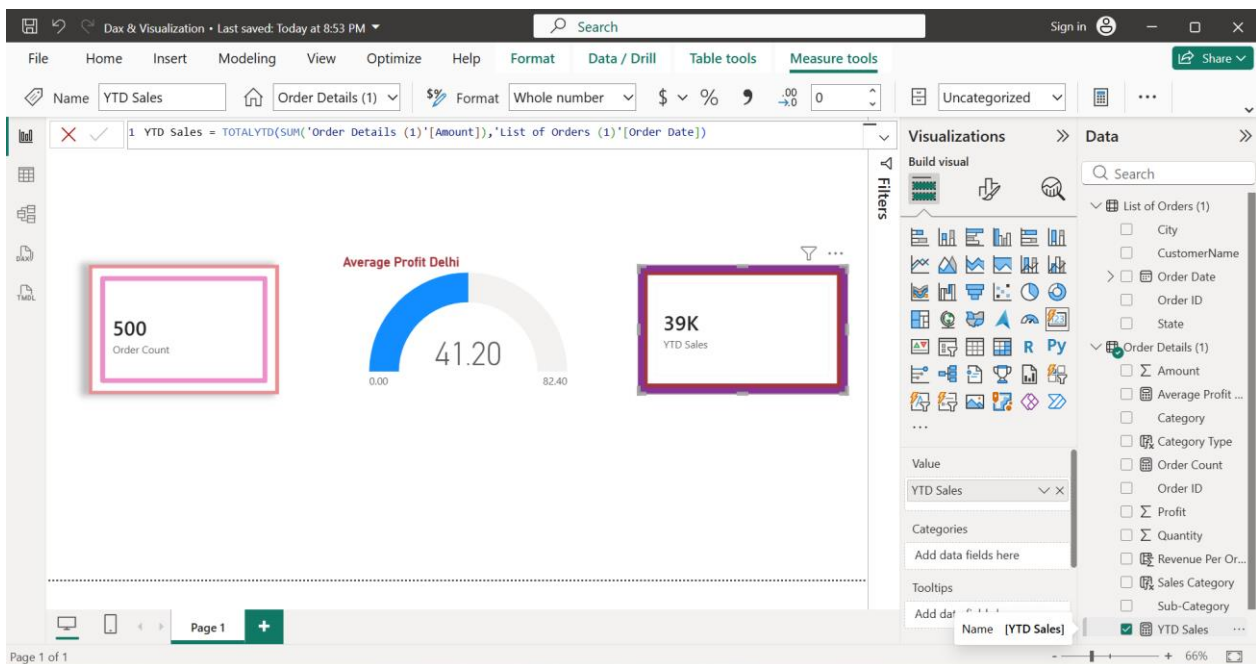
Revenue Per Order

Sales Category

Sub-Category

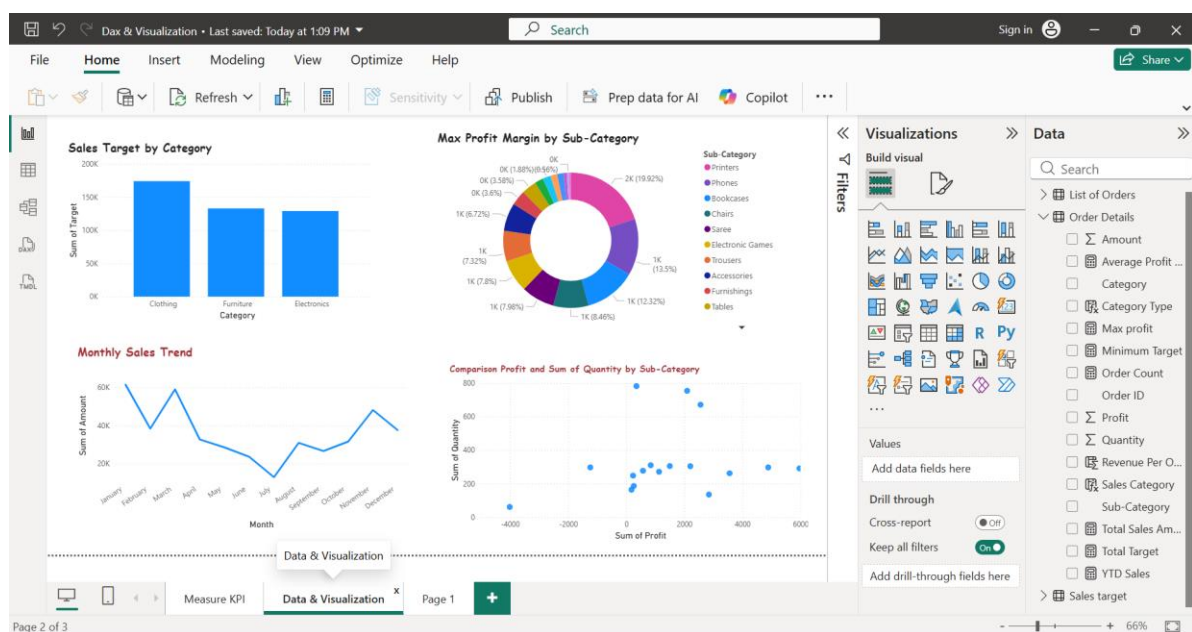
All tables

Calculated Measures / KPI:



- ◆ Order Count = Count ('Order Detail' [Order ID])
- ◆ Average Profit in Delhi = CALCULATE(Average('Order Details'[Profit]),'List of Orders'[City]= "Delhi")
- ◆ YTD Sales = TOTALYTD(SUM('Order Details'[Amount],'List of Orders'[Order Date])

Data Visualization:



1. Sales Target Achievement by Category

Using a Clustered Column Chart : X Axis →Category, Y Axis →Sales Target

2. Max Profit Margin by Sub-Category

Using a Donut Chart : Legend →Sub-Category, Values →Max Profit

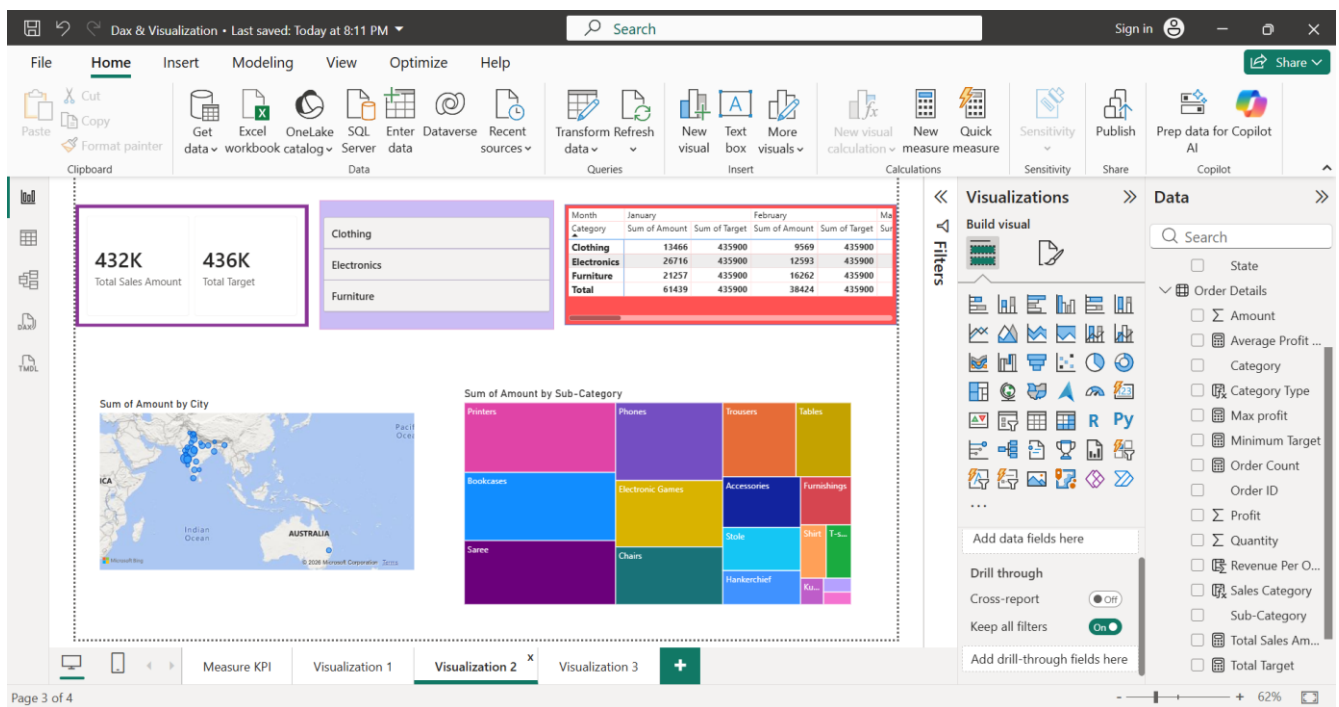
3. Monthly Sales Trend

Using a Line Chart : X Axis →Order Date(Month), Y Axis →Sum of Amount

4. Comparison of Profit and Quality by Sub-Category:

Using a Scatter Chart: X Axis→Sum of Quantity, Y Axis →Sum of Profit

Values →Sub-Category



5. Comparison of Total Sales Amount and Target: Using Card Visual

6. Sales Performance Matrix: Row →Category, Columns→Order Date(Month)

Values→Sum of Amount, Sum of Target

7. Geographic Sales Analysis: Using Visualize Map

8. Sales Distribution by Sub Category: Using a Treemap Visual

Category → Sub-Category, Values → Sum of Amount

9. Order Count Analysis by State: Create a Funnel Chart

Category → State, Values → Order Count

